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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous College under V T U, Belgaum)

Seventh Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE702/20EE703/17EE703 - INDUSTRIAL DRIVES AND APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT	CO	PO
1.	a) With neat block diagram explain the basic components of an electric drive.	7	L*2	1	1,2
	b) With sign convention explain the multiquadrant operation of a motor driving a hoist load.	8	L3	1	1,2
	c) Obtain the equivalent values of drive parameters for loads with translational motion.	5	L2	1	1,2
2.	a) Derive an expression for overloading factor K when the motor is subjected to intermittent load. Draw the necessary waveforms.	8	L2	2	1,2
	b) Obtain the expression for moment of inertia of a flywheel connected for load equalization.	7	L3	1	1,2
	c) A constant speed drive has the following duty cycle. (i) Load rising from 0 to 400kW for 5 min (ii) Uniform load of 500kW for 5 min (iii) Regenerative power of 400kW returned to supply for 4 min (iv) remains idle for 2 min Estimate the power rating of motor. Assume losses to be proportional to square of the power.	5	L3	1	1,2
3.	a) Analyze the various classes of motor duty with diagrams.	8	L2	2	1,2
	b) A 220V, 970rpm, 100A dc separately excited motor has an armature resistance of 0.05Ω. It is braked by plugging from an initial speed of 1000rpm. Calculate (i) resistance to be placed in armature circuit to limit the braking current twice the full load value (ii) braking torque (iii) torque when the speed has fallen to zero.	7	L3	2	1,2
	c) With torque speed characteristics explain the dynamic braking of separately excited DC motor.	5	L2	2	1,2
Unit – II					
4.	a) With neat circuit diagram and waveforms analyze the operation of single phase fully controlled rectifier fed dc separately excited motor for continuous mode of conduction.	10	L2	3	1,2
	b) A 200V, 875rpm, 150A separately excited DC motor has an armature resistance of 0.06Ω. It is fed from a single phase fully controlled rectifier with an AC source of 220V, 50Hz. Assuming continuous conduction, calculate (i) firing angle for rated motor torque and 750rpm (ii) firing angle for rated motor torque and -500rpm (iii) motor speed for $\alpha=160^\circ$ and rated torque. Assuming dynamic braking resistor of 2Ω connected while braking.	10	L3	3	1,2
5.	a) With neat circuit explain the chopper controlled dc separately excited motor for motoring and Regenerative braking.	10	L2	3	1,2
	b) A 230V, 960rpm, 200A dc separately excited motor has an armature resistance of 0.02Ω. The motor is fed from a chopper which provides both motoring and braking operations. The source has a voltage of 230V. Assuming continuous conduction, calculate (i) duty ratio of chopper for a motor speed of 600rpm and braking torque of twice the rated value. (ii) what will be the motor speed for a duty ratio of 0.6 and motor torque equal to twice the rated value.	10	L3	3	1,2

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| 6. | a) | Explain the operation of voltage source inverter fed induction motor drive. Also sketch various schemes of VSI fed induction motor drives. | 10 | L2 | 4 | 1,2 |
| | b) | With neat circuit and speed torque curves explain the static Scherbius drive scheme for slip power recovery. | 10 | L3 | 4 | 1,2 |

Unit – III

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| 7. | a) | Explain the operation of self-controlled synchronous motor drive employing load commutated thyristor inverter. | 10 | L2 | 5 | 1 |
| | b) | Write a note on (i) rolling mill drives (ii) Paper mill drives | 10 | L1 | 5 | 1 |
| 8. | a) | Explain the operation of a synchronous motor when fed from fixed frequency supply. | 10 | L2 | 5 | 1 |
| | b) | With neat schematic diagram explain the process involved and motors used in cement mill drives. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
